



Foundations of Programming

Assignment No.: 21

PROBLEM STATEMENT: Write a C program to copy contents of one file to another using file handling.

OBJECTIVE:

- To understand the concept of file handling in the C programming language.
- To learn how to read data from a file and write data into another file.
- To implement file copy operation using standard file handling functions.

THEORY:

File handling in the C programming language allows programs to store and retrieve data from files. Files are handled using file pointers and standard functions such as `fopen()`, `fclose()`, `fgetc()`, and `fputc()`.

In this program, the source file is opened in read mode, and the destination file is opened in write mode. The program reads characters from the source file one by one using `fgetc()` and writes them to the destination file using `fputc()`. This process continues until the end of the file (EOF) is reached. Thus, the contents of the source file are successfully copied into the destination file.

PLATFORM: Linux

INPUT: A source file containing text data.

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      FILE *source, *destination;
6      char ch;
7
8      // Open source file in read mode
9      source = fopen("source.txt", "r");
10     if (source == NULL) {
11         printf("Error opening source file.\n");
12         return 1;
13     }
14
15     // Open destination file in write mode
16     destination = fopen("destination.txt", "w");
17     if (destination == NULL) {
18         printf("Error opening destination file.\n");
19         fclose(source);
20         return 1;
21     }
22
23     // Copy contents character by character
24     while ((ch = fgetc(source)) != EOF) {
25         fputc(ch, destination);
26     }
27
28     // Close files
29     fclose(source);
30     fclose(destination);
31
32     printf("File copied successfully.\n");
33
34     // Display contents of destination file
35     destination = fopen("destination.txt", "r");
36     if (destination == NULL) {
37         printf("Error reading destination file.\n");
38         return 1;
39     }
40
41     printf("Destination file destination.txt will contain:\n");
42     while ((ch = fgetc(destination)) != EOF) {
43         printf("%c", ch);
44     }
45
46     fclose(destination);
47
48     return 0;
49 }

```

OUTPUT: Contents of the source file copied into another file.

```
File copied successfully.  
Destination file destination.txt will contain:  
Hello  
Welcome to C Programming
```

SAMPLE INPUT:

source.txt

Hello

Welcome to C Programming

SAMPLE OUTPUT:

File copied successfully.

Destination file **destination.txt** will contain:

Hello

Welcome to C Programming

CONCLUSION: Thus, the C program successfully demonstrates **file handling operations** by copying the contents of one file into another using file handling functions.

FAQs:

1. What is **file handling in C programming**, and why is it important?
2. What is the purpose of **fopen() and fclose() functions** in file operations?
3. How does the program detect the **end of a file using EOF**?